

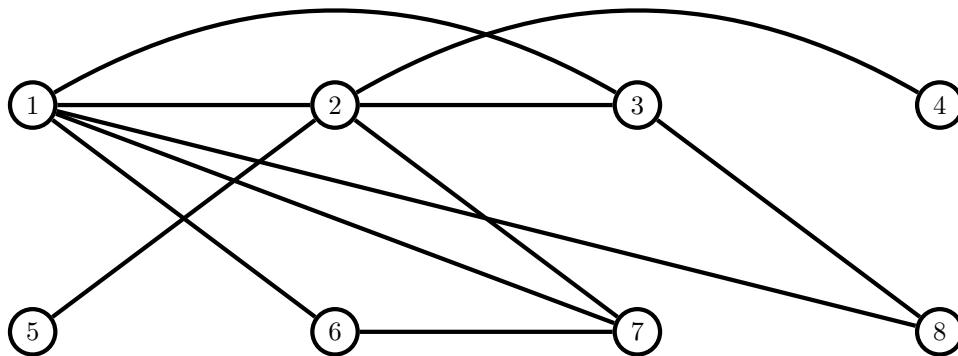
Tutorial 9

Exercise 1: Modelling

1. Model knapsack as an optimization problem according to the definition on Slide 6 of Lecture 9.
2. Model vertex cover as an optimization problem according to the definition on Slide 6 of Lecture 9.

Exercise 2: Vertex cover

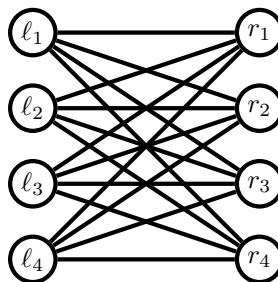
1. Apply the vertex cover approximation algorithm to the following graph.



2. The complete bipartite graph $K_{m,m}$ is defined as $K_{m,m} = (V_m, E_m)$ with

- $V_m = \{\ell_j, r_j \mid 1 \leq j \leq m\}$ and
- $E_m = \{(\ell_j, r_{j'}), (r_j, \ell_{j'}) \mid 1 \leq j, j' \leq m\}$.

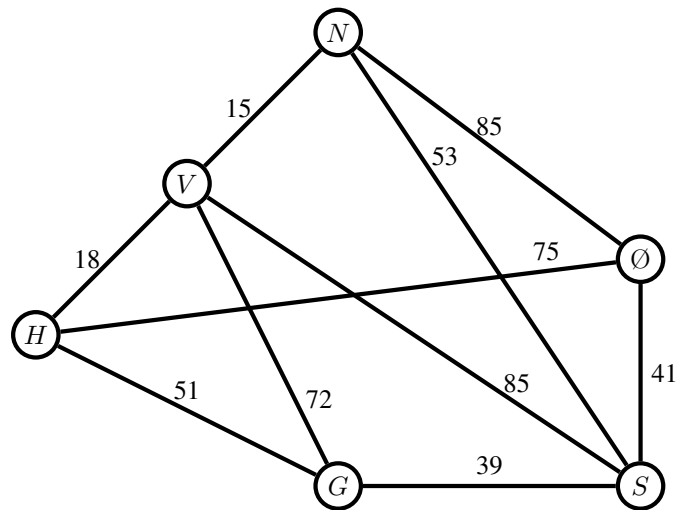
Intuitively, $K_{m,m}$ consists of two disjoint sets of m vertices, and all edges between these sets, but no edges in the two sets. $K_{4,4}$ is depicted below.



- (a) What is the size a smallest vertex cover in $K_{m,m}$?
- (b) What is the size of a vertex cover computed by the approximation algorithm presented during the lecture?
- (c) What does this tell us about the approximation ratio of the algorithm?

Exercise 3: Minimum spanning trees

Compute an MST for the following graph using Prim's algorithm.

**Exercise 4: Travelling Salesperson**

Apply the approximation algorithm for TSP to the graph from Exercise 3.

Exercise 5: Decision vs. optimization problems

In the lecture, we have claimed that “Decision and optimization are interreducible with polynomial overhead”.

1. Formalize the statement for the decision and the optimization variant of the clique problem.
2. Prove your statement.
3. Can you think of a situation where the interreducibility fails?